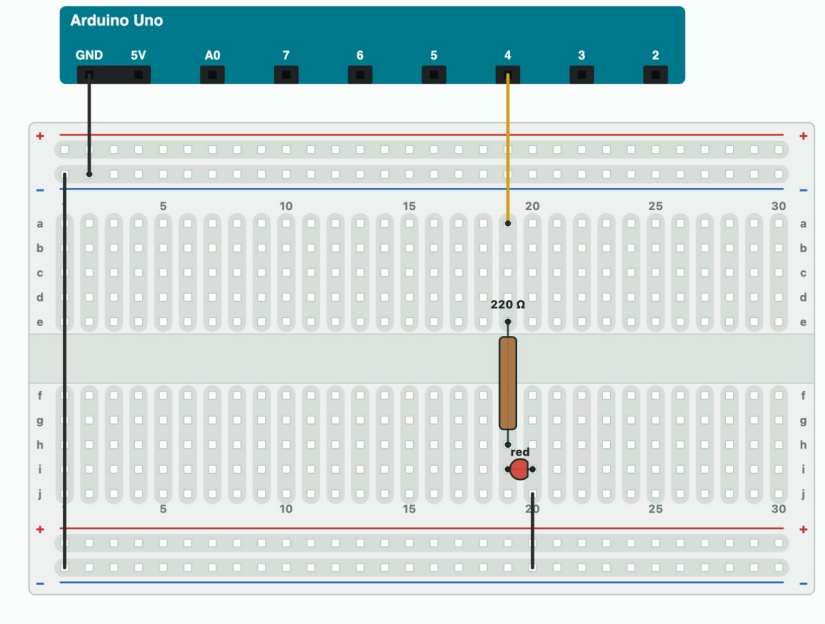
Daily Review



1	delay(1000) waits…	1 s
2	setup() runs how often?	Once
3	loop() runs…	For ever
4	digitalWrite(4, HIGH); does…	Switches pin 4 on
5	The exam's pause 500 is…	0.5 s

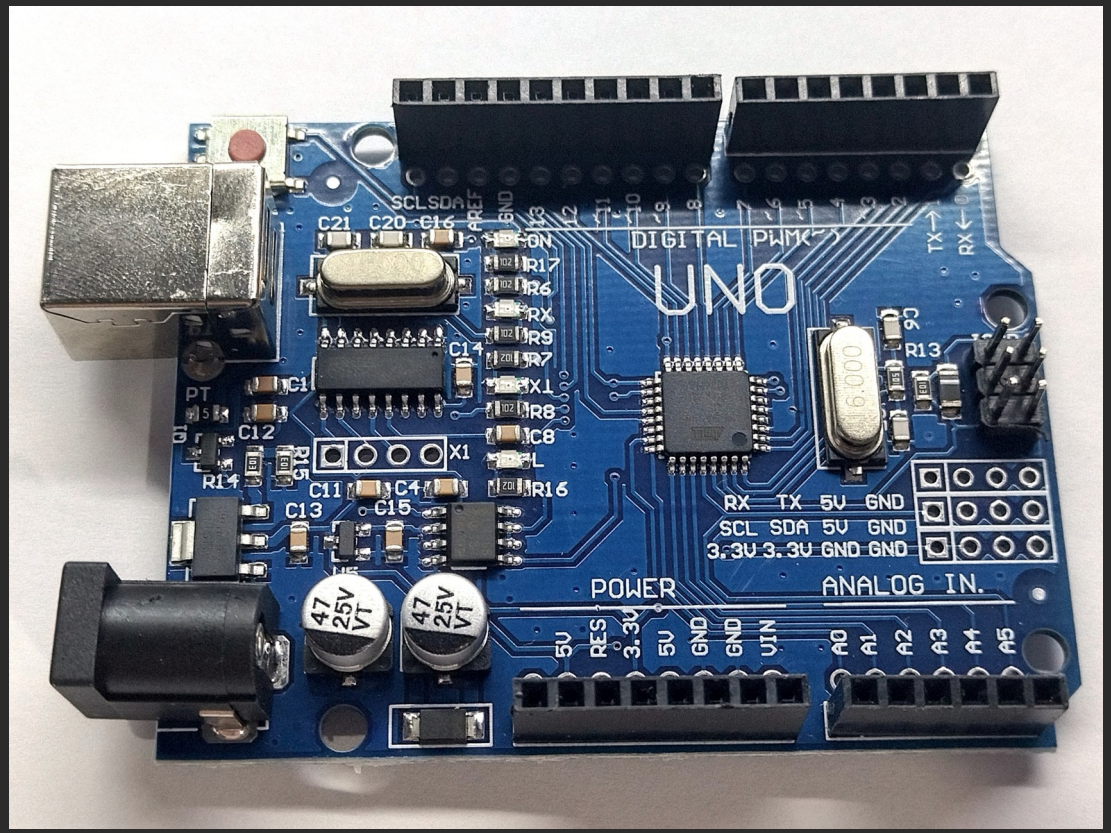
Build, Test and Fix

Put the program on a real Uno. Then make it work.



National 5 Engineering Science

This board has no program on it yet. What must you give it to flash an LED?



An Arduino Uno

PART 1

Build It

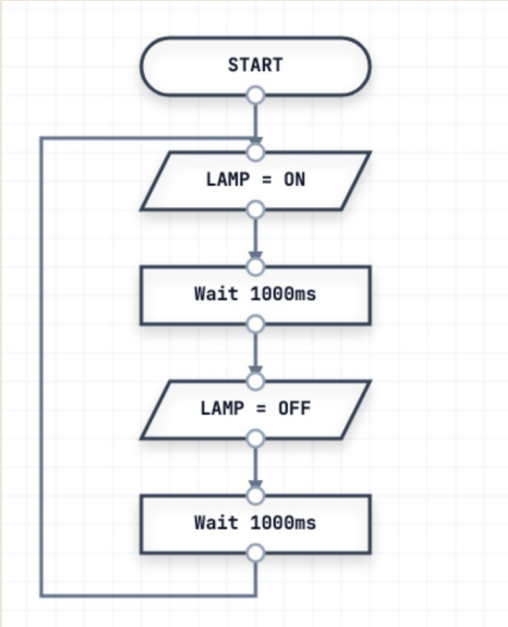
Simulate, verify, upload, build.

By the end of this part…

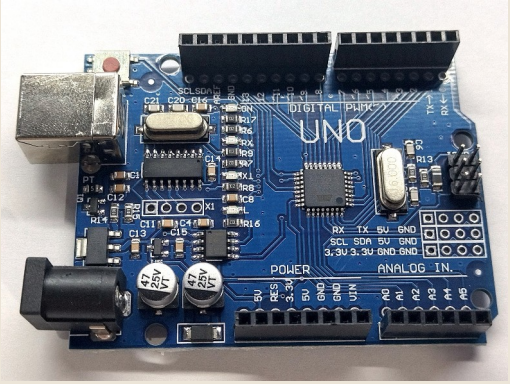


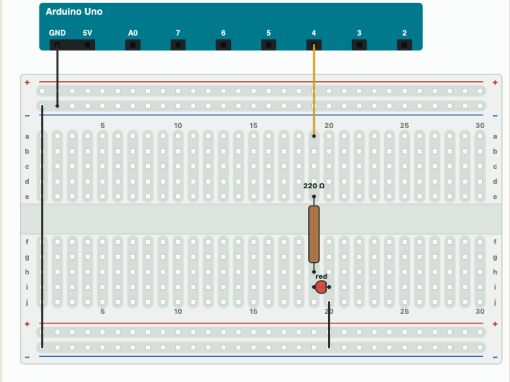
1. I can wire an LED to a pin on the Uno
2. I can match each line of code to a flowchart box
3. I can upload a program and check it works

From Sim to Board



Generate
Code
Duino.app
Verify





1 Sim

2 Verify

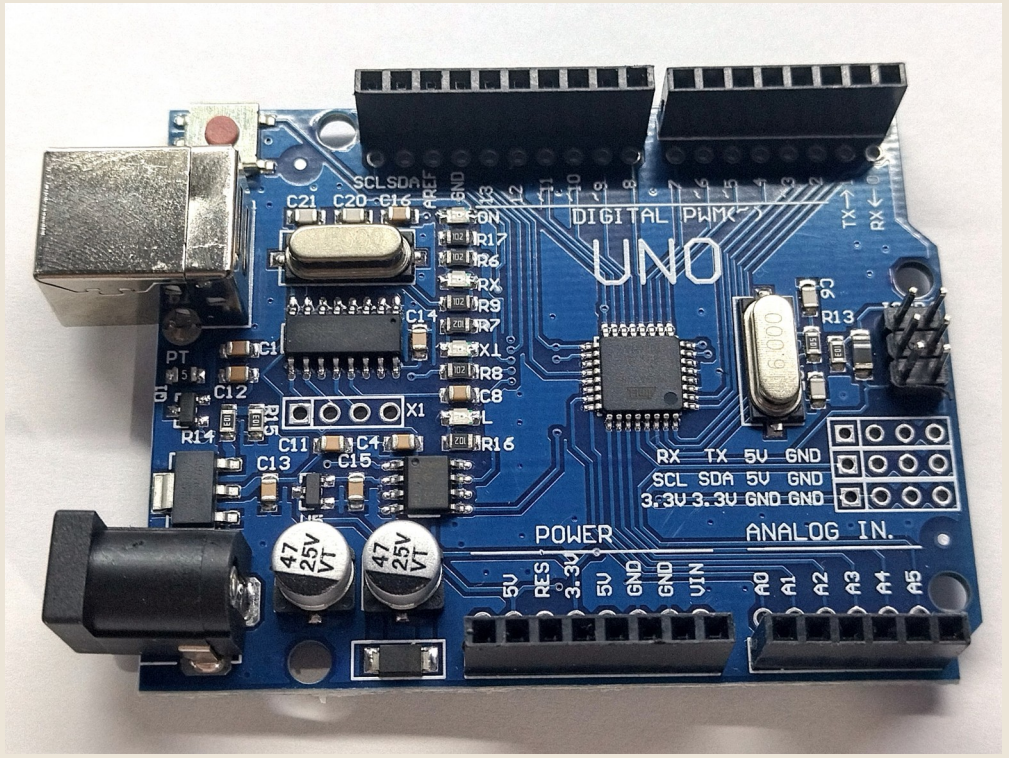
3 Upload

4 Build

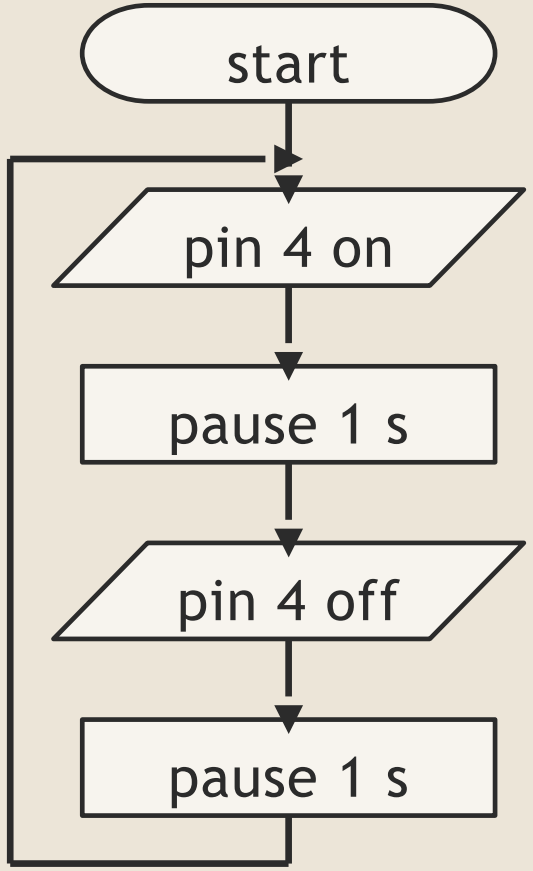
What to Collect

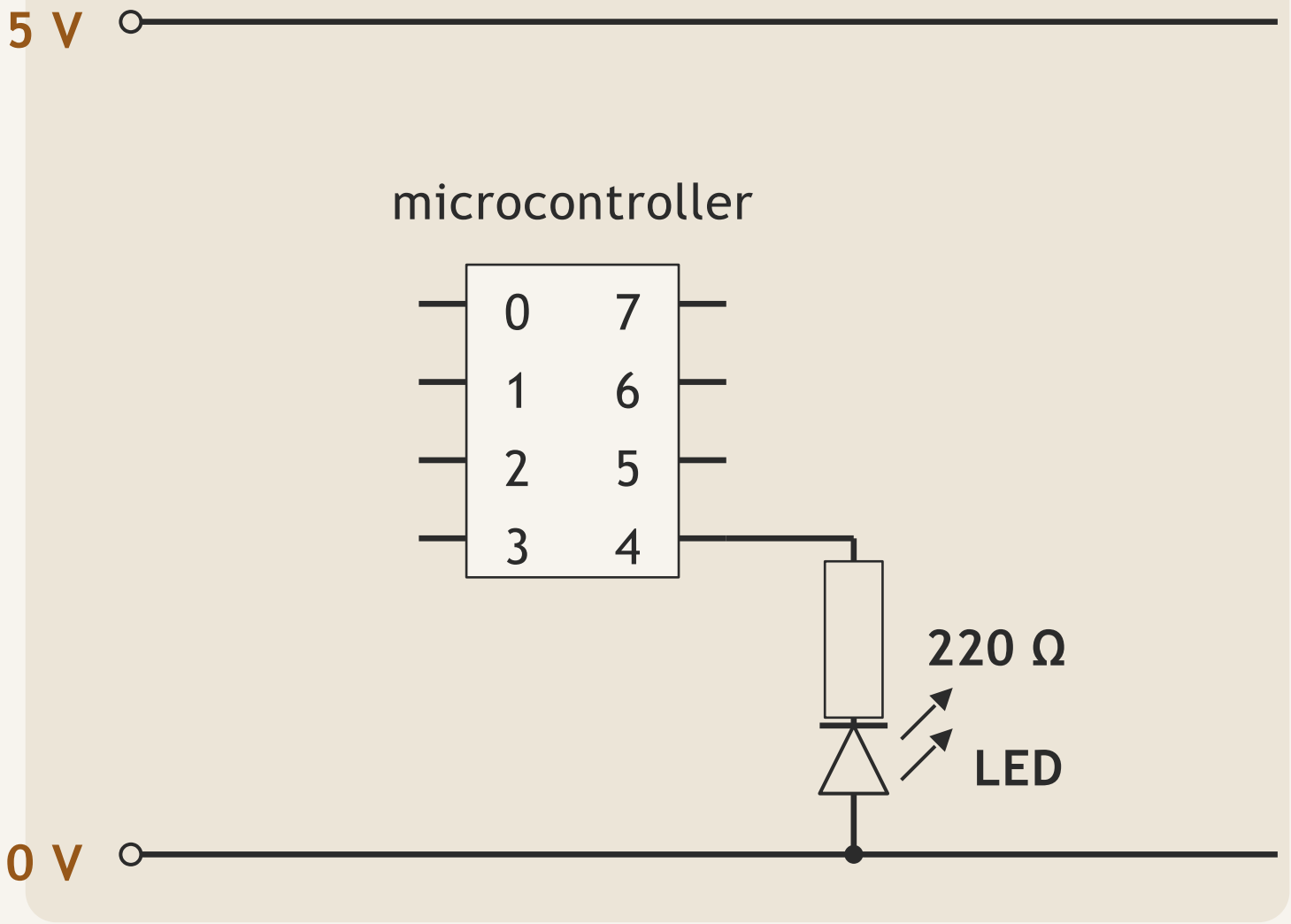
Collect one set per pupil.

1 Uno and its USB lead
2 Breadboard
3 3 LEDs: red, amber, green
4 3 × 220 Ω resistors
5 6 jumper leads



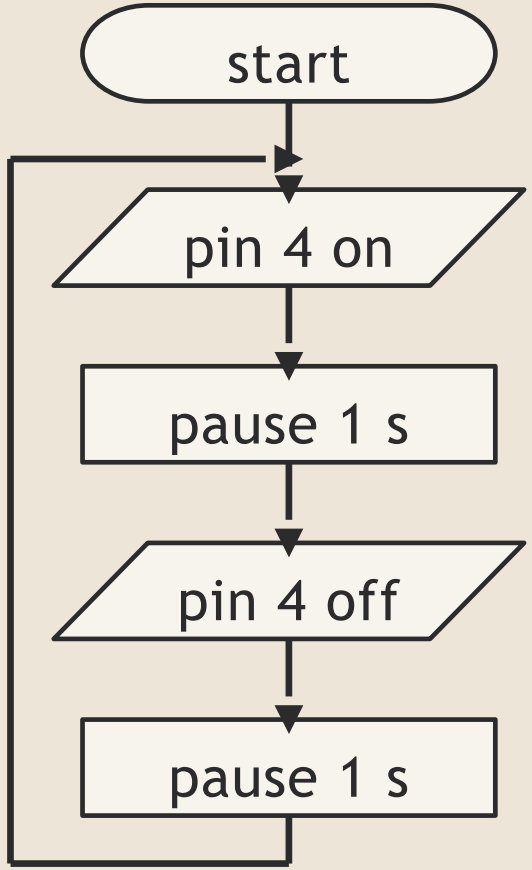
Worked Example 1: The Plan and the Wires



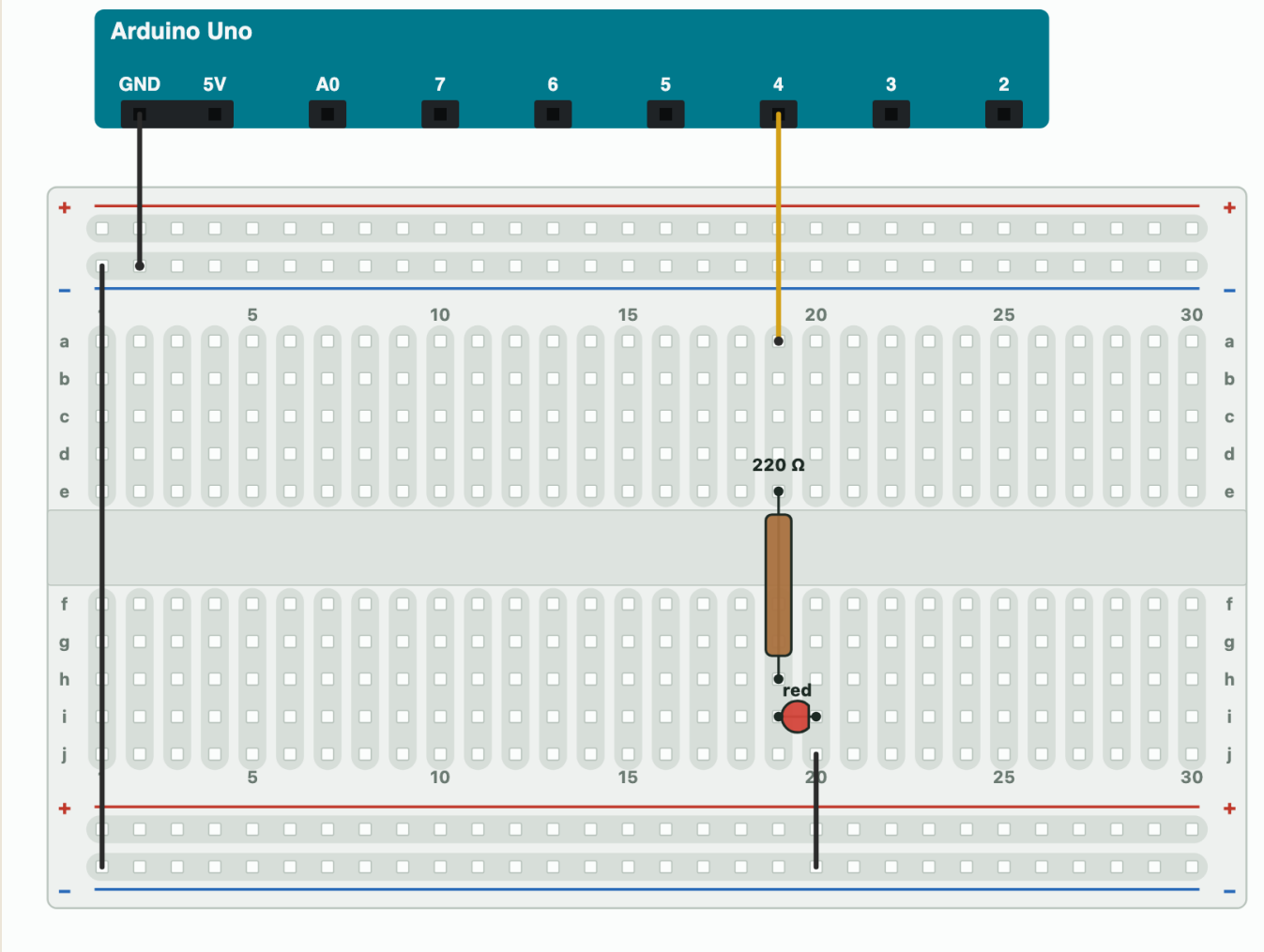


Worked Example 1: The Code

```
1   void setup() {
2     pinMode(4, OUTPUT);
3   }
4   void loop() {
5     digitalWrite(4, HIGH);
6     delay(1000);
7     digitalWrite(4, LOW);
8     delay(1000);
9   }
```



Worked Example 1: Build It



Yellow lead

pin 4 to the 220 Ω

LED

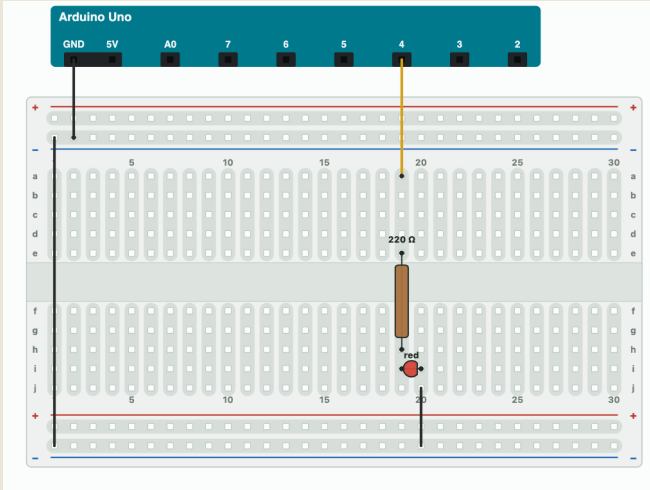
long leg to the resistor

Black lead

GND to the – rail

Build It: Worked Example 1

Practical sheet 1



NoSyntaxSim → Duino.app → Verify.
Upload, unplug, wire it. 1 s on, off?

SAFETY USB power only. Unplug before you move a wire.

15 min

Try Changing This: Worked Example 1



1. Both delays 200. Faster or slower?

Faster: 2.5 flashes a second

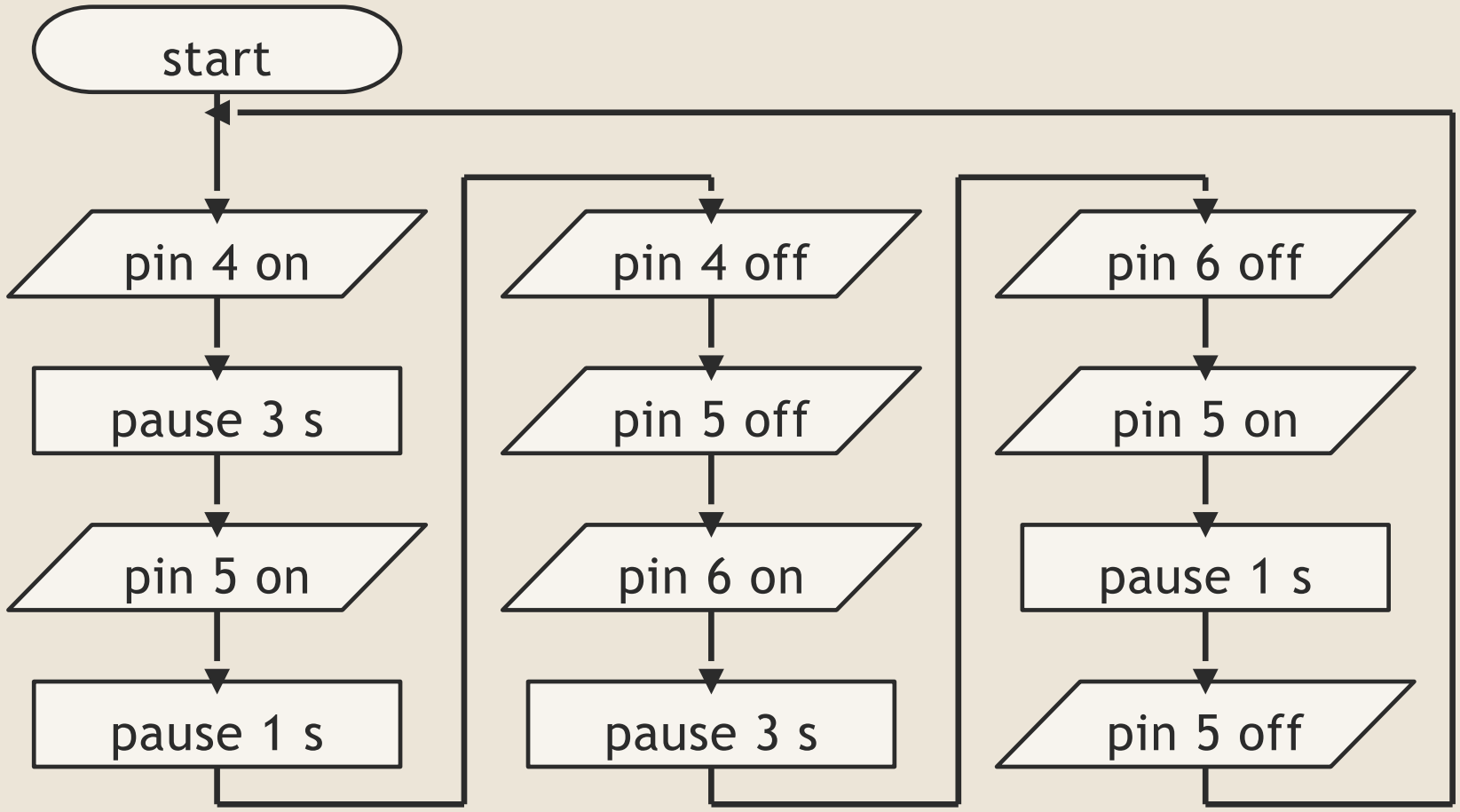
2. delay(100), then delay(900)?

A short blink once a second

3. Move the LED to pin 5. What else changes?

All three pin 4s — code must match the wire

Worked Example 2: Traffic Lights



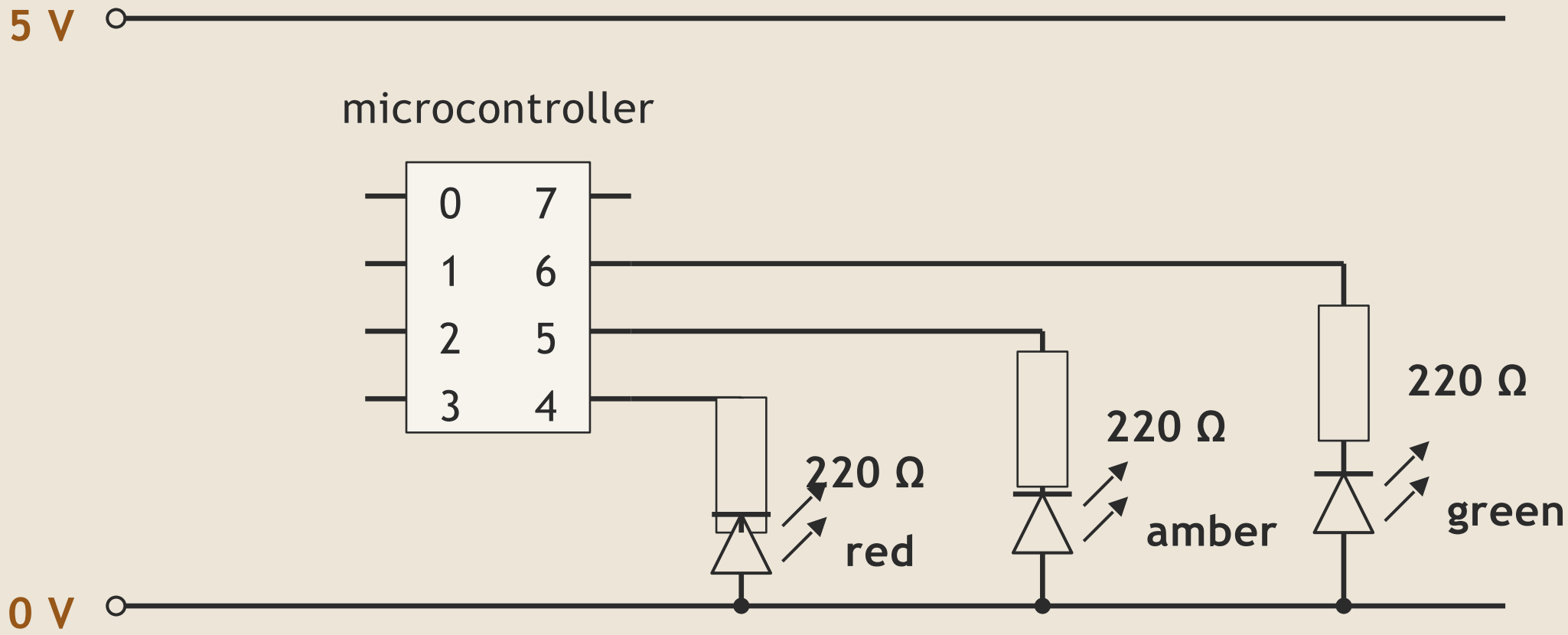
1 red

2 red + amber

3 green

4 amber

Worked Example 2: The Wires



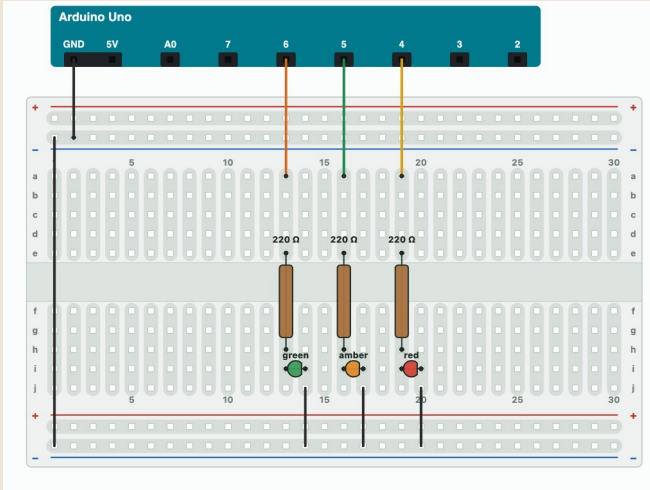

```
pinMode(4, OUTPUT);  pinMode(5, OUTPUT);  pinMode(6, OUTPUT);
```

Worked Example 2: The Code

```
1   void loop() {
2     digitalWrite(4, HIGH);
3     delay(3000);
4     digitalWrite(5, HIGH);
5     delay(1000);
6     digitalWrite(4, LOW);
7     digitalWrite(5, LOW);
8     digitalWrite(6, HIGH);
9     delay(3000);
10    digitalWrite(6, LOW);
11    digitalWrite(5, HIGH);
12    delay(1000);
13    digitalWrite(5, LOW);
14  }
```

Build It: Worked Example 2

Practical sheet 1



NoSyntaxSim → Duino.app → Verify.
Upload, unplug, add amber and green.

SAFETY USB power only. Unplug before you move a wire.

15 min

Try Changing This: Worked Example 2



1. Green on for 6 s, not 3. Which line?

   Line 9: delay(6000)

2. Add // by your name. Does it still run?

   Yes — // is a comment

3. Swap the 4 and 6 in setup() only?

   Nothing — all three are still outputs

Checkpoint



1. Why the 220 Ω resistor?

   Stops too much current — protects the LED

2. A pin is not set in setup(). What happens?

   LED off, or only glows dimly

3. Two LEDs on at once in a flowchart?

   Two output boxes, one after the other

Plenary: Part 1



1	setup() sets each pin as…	Input or output
2	The closing } of loop() acts like…	The arrow back up
3	delay(3000) waits…	3 s
4	The code's pin must match…	The wire
5	Verify, then…	Upload

Look back — by the end of this part…



1. I can wire an LED to a pin on the Uno
2. I can match each line of code to a flowchart box
3. I can upload a program and check it works

Daily Review



1	pinMode(5, OUTPUT); goes in…	setup()
2	delay(250) waits…	0.25 s
3	Red and amber on: how many boxes?	Two
4	Why a 220 Ω with each LED?	To protect it
5	The code uses pin 5, the wire pin 4?	Nothing happens

PART 2

Sensors and Faults

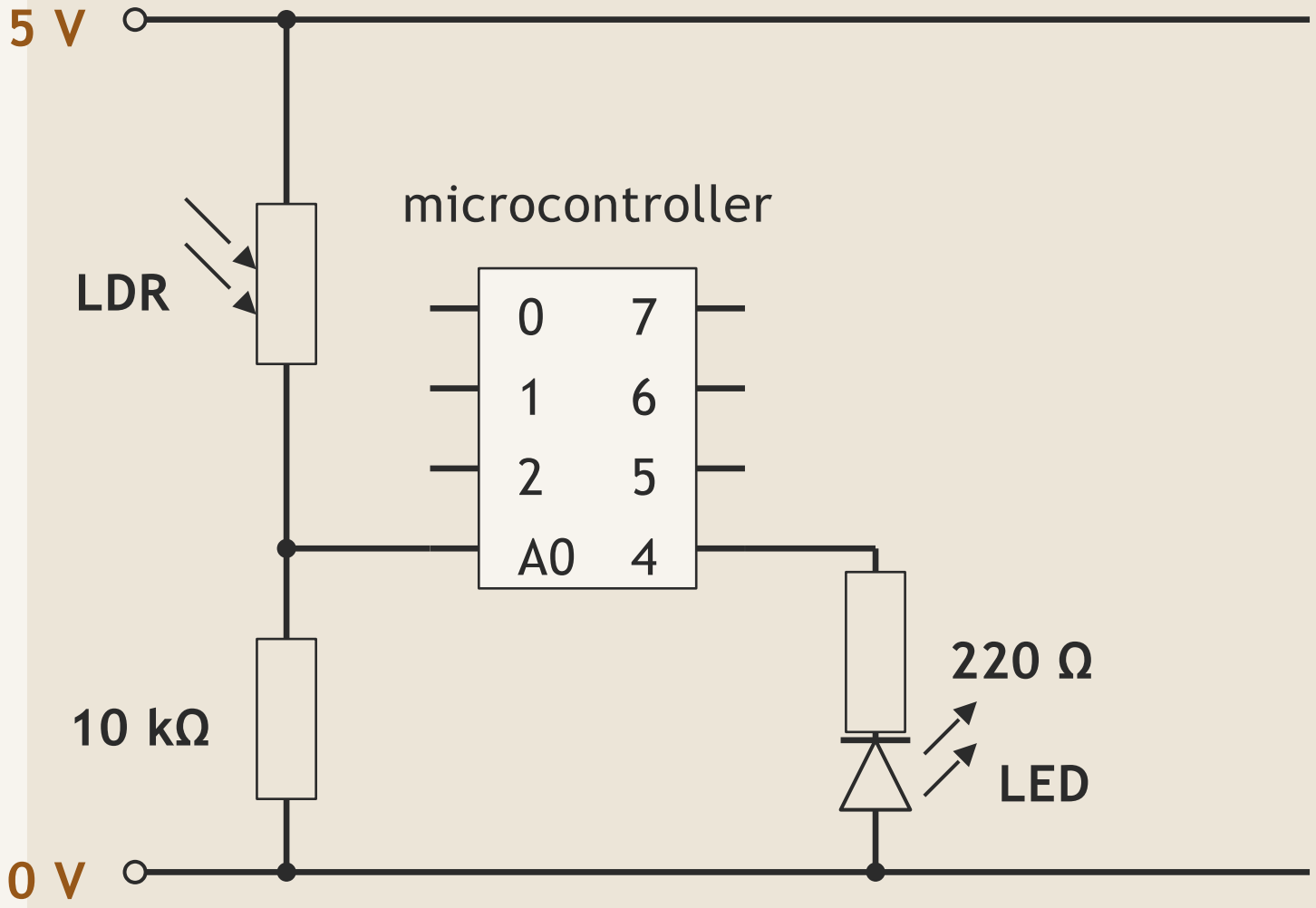
Make it see the dark. Then find what is wrong.

By the end of this part…



1. I can read a sensor with analogRead()
2. I can count a fixed loop in Arduino C
3. I can find and fix faults in a program

Worked Example 3: The LDR on A0



analogRead(A0)

a number from 0 to 1023

Dark

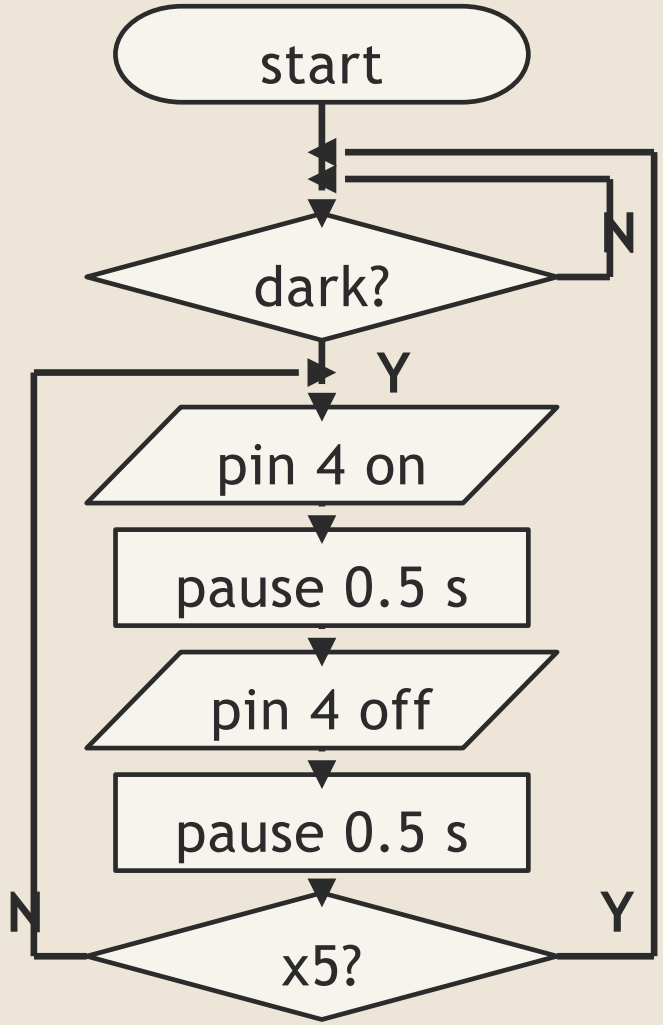
LDR resistance up, A0 down

light < 300

means it is dark

Worked Example 3: Flowchart and Code

```
1   int light = analogRead(A0);
2   if (light < 300) {
3     for (int i = 0; i < 5; i++) {
4       digitalWrite(4, HIGH);
5       delay(500);
6       digitalWrite(4, LOW);
7       delay(500);
8     }
9   }
```



Counting From 0




```
for (int i = 0; i < 5; i++)
```

i = 0

start the count at 0

i < 5

keep going while i is under 5

i++

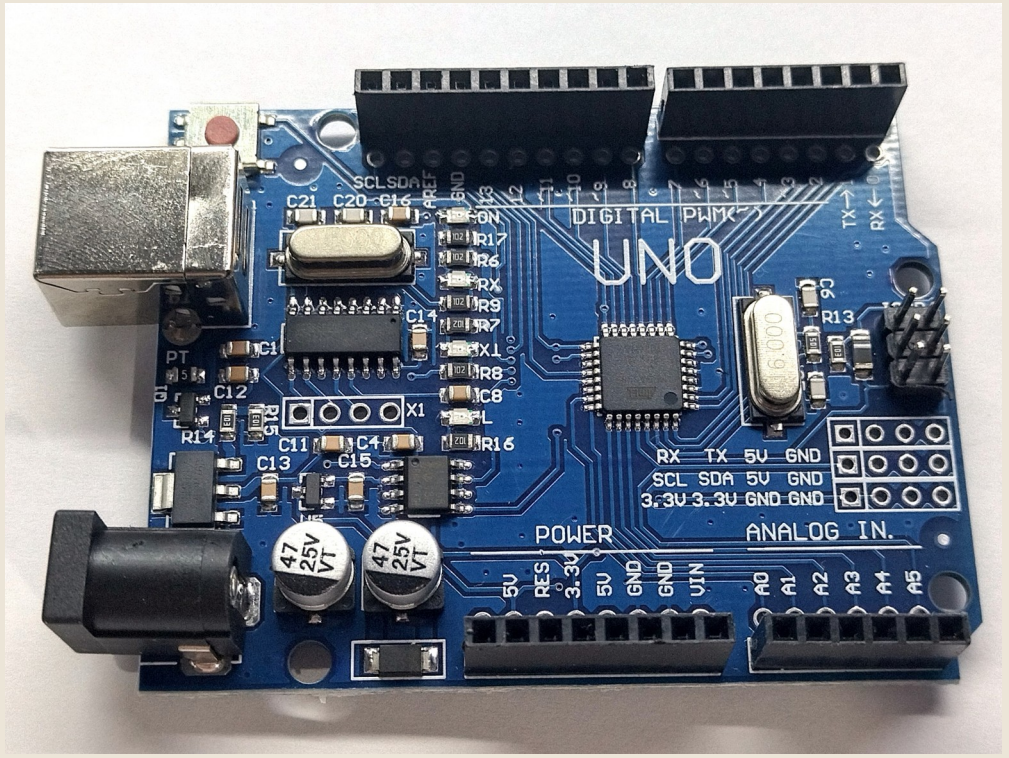
add 1 after each pass

i	i < 5?	flash
0	yes	1
1	yes	2
2	yes	3
3	yes	4
4	yes	5
5	no	stop

What to Collect

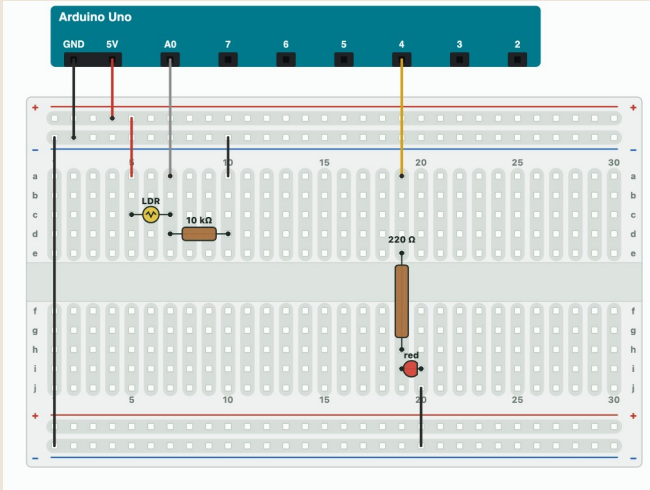
Collect one set per pupil.

1 LDR
2 2 × 10 kΩ (brown, black, orange)
3 Push button
4 Buzzer
5 4 more jumper leads



Build It: Worked Example 3

Practical sheet 2



NoSyntaxSim → Duino.app → Verify.
Upload, unplug, add the LDR (page 18).

SAFETY USB power only. Unplug before you move a wire.

15 min

Try Changing This: Worked Example 3



1. Change 300 to 600. Easier or harder?

Easier — it counts as dark sooner

2. i < 10. How many flashes?

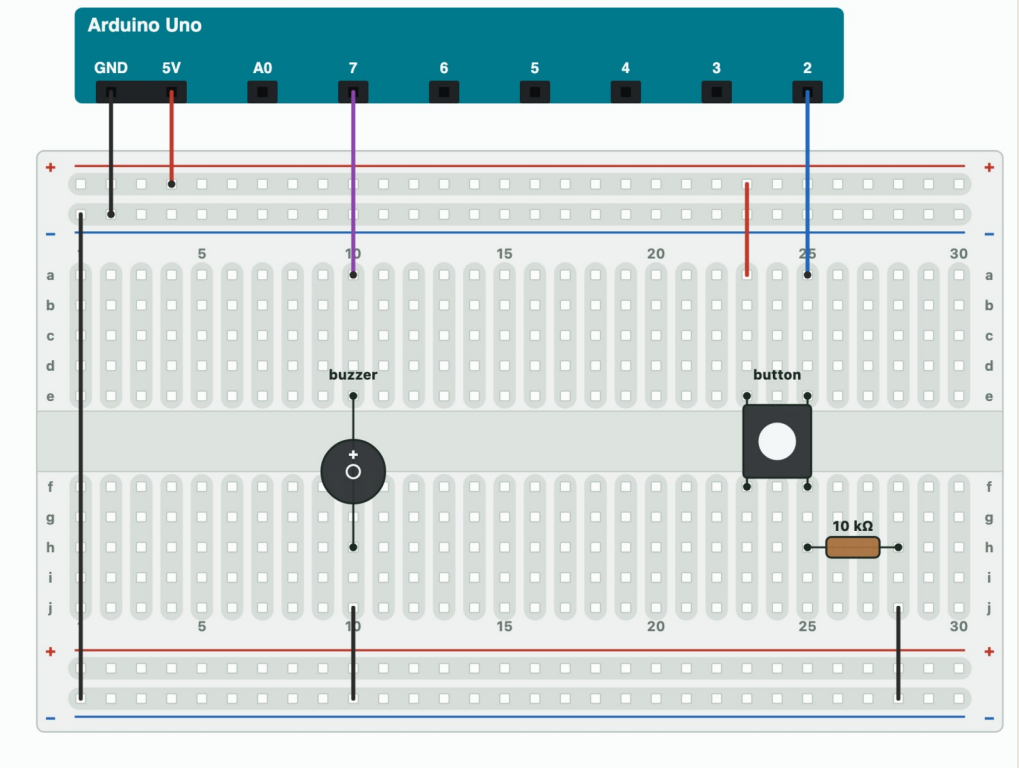
10

3. Swap < for >. When does it go off?

In the light

Debugging: The Method

1. What should it do?
2. What does it do?
3. Which line?
4. Change one thing
5. Test it again



The Faults to Look For

No pinMode

the pin is never made an output

delay(1)

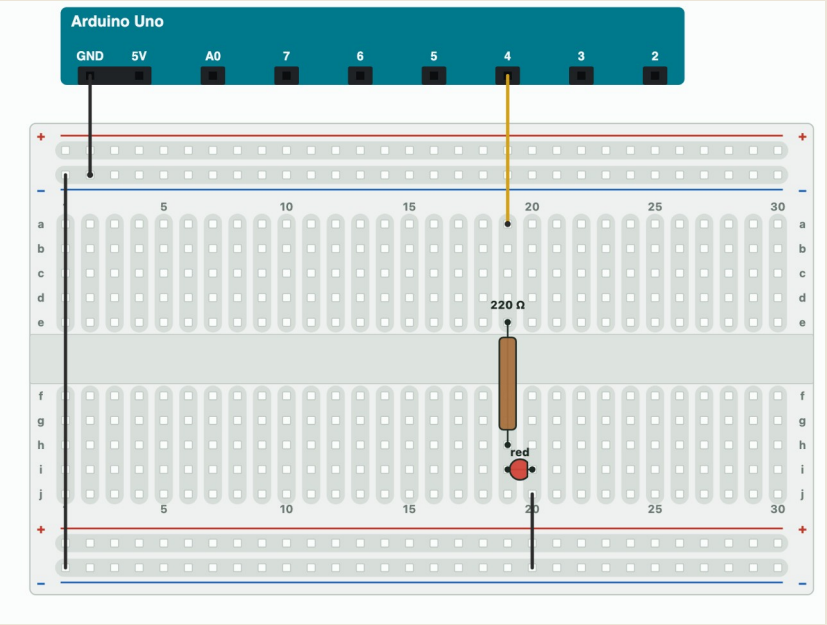
1 ms, not 1 s

= in an if

compare with ==

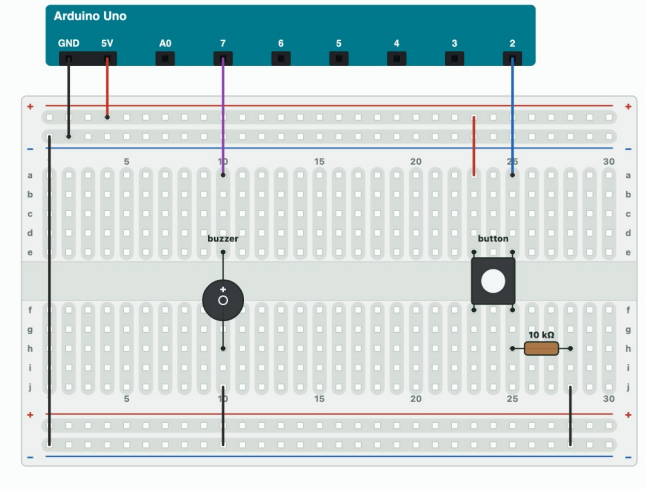
A missing }

every { needs its }



Find and Fix: Debug Builds 1 and 2

Practical sheet 2



Duino.app: type each in, Verify, Upload.
Wire it. Find 2 faults in each. Re-test.

SAFETY USB power only. Unplug before you move a wire.

15 min

Find and Fix: Debug Build 1



```
1   void setup() {
2   }
3   void loop() {
4     digitalWrite(4, HIGH);
5     delay(1);
6     digitalWrite(4, LOW);
7     delay(1000);
8   }
```

Should: flash pin 4, 1 s on, 1 s off.

1 Line 2

Add pinMode(4, OUTPUT);

2 Line 5

delay(1000);

Find and Fix: Debug Build 2



```
1   void setup() {
2     pinMode(2, INPUT);
3     pinMode(7, OUTPUT);
4   }
5   void loop() {
6     if (digitalRead(2) = HIGH) {
7       digitalWrite(7, HIGH);
8       delay(2000);
9       digitalWrite(7, LOW);
10    }
```

Should: buzz 2 s when the button is pressed.

1 Line 6

== HIGH, not = HIGH

2 Line 11

Add } to close loop()

Watch Out: The Exam Never Asks for Code

THE MISTAKE

- digitalWrite(4, HIGH);
- delay(1000);
- Code in an exam answer.

THE CORRECT ANSWER

- pin 4 on
- pause 1 s
- A pin number and a unit.

Checkpoint



1. analogRead(A0) gives a number from…

   0 to 1023

2. for (i = 0; i < 3; i++) runs how often?

   3 times

3. Why change one thing at a time?

   So you know which change fixed it

Plenary: Part 2



1	In the dark, A0's reading goes…	Down
2	== or = to compare?	==
3	Every { needs…	Its }
4	The fixed loop in the flowchart is…	x5? (a decision)
5	In the exam, never write…	Arduino C

Look back — by the end of this part…



1. I can read a sensor with analogRead()
2. I can count a fixed loop in Arduino C
3. I can find and fix faults in a program

Daily Review



1	delay(1) waits…	1 ms
2	= or == in an if?	==
3	A0 needs a pinMode line?	No
4	Debugging step 4?	Change one thing
5	The exam's answer is never in…	Arduino C

PART 3

Test, Amend, Re-test

It works once. Does it do every job?

By the end of this part…



1. I can write a test for each job a program does
2. I can say what happened, not just 'it failed'
3. I can plan a change and test it again

Why Engineers Test

Job 1

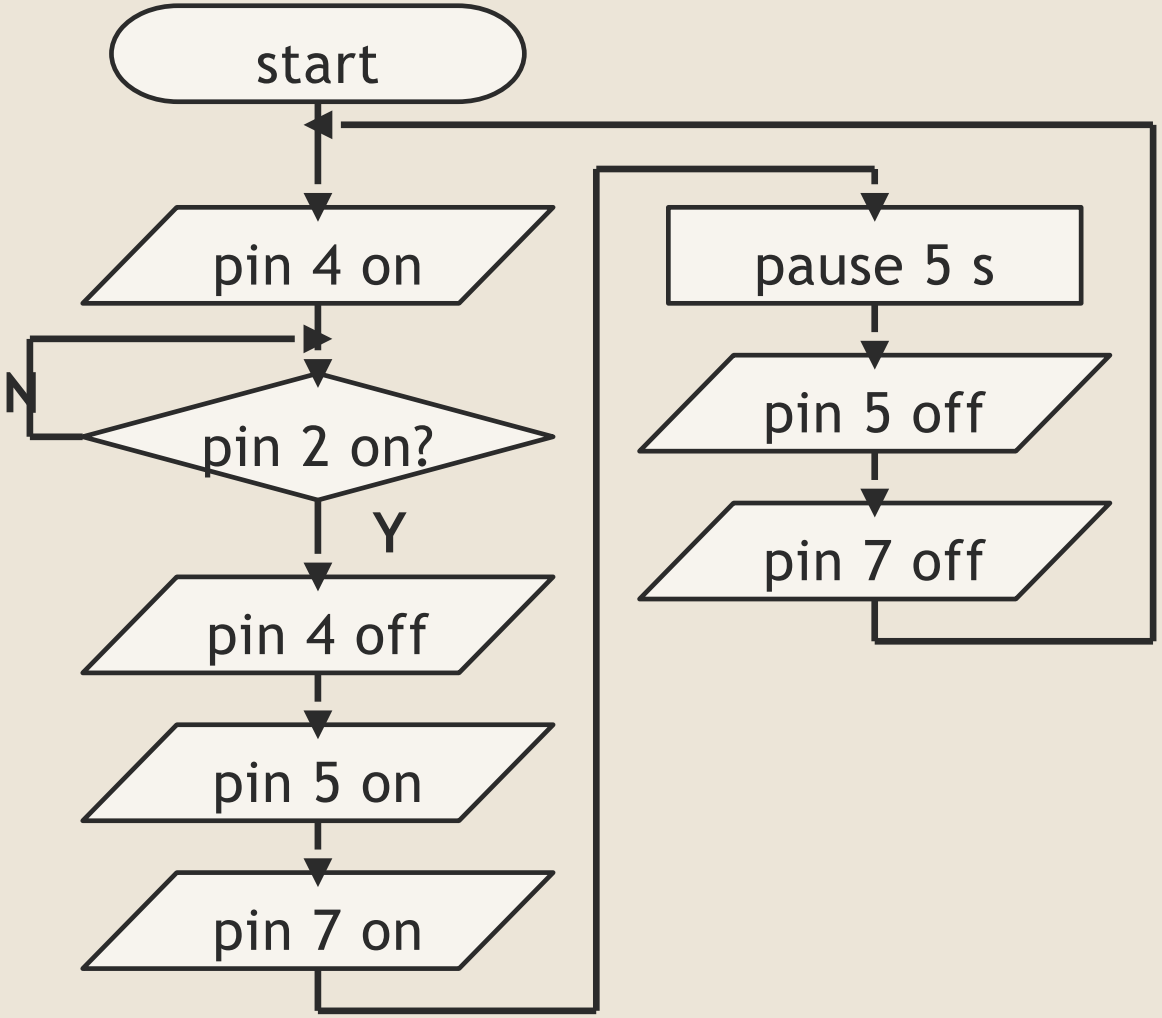
red until the button is pressed

Job 2

green and beeper for 5 s

Job 3

back to red, beeper off



The Test Table



Test	Should happen	Initial result	Planned amendment	Re-test

Initial result: what did happen · Amendment: the change

Worked Example: The Crossing

Test	Should happen	Initial result	Planned amendment	Re-test
Press	Red off, green on	Red off, green on	None	
Wait 5 s	Green off, red on	Green off, red on	None	
Beeper	Stops after 5 s	Stays on	Add pin 7 off	Stops: pass

The Fault, in the Sim

Last box

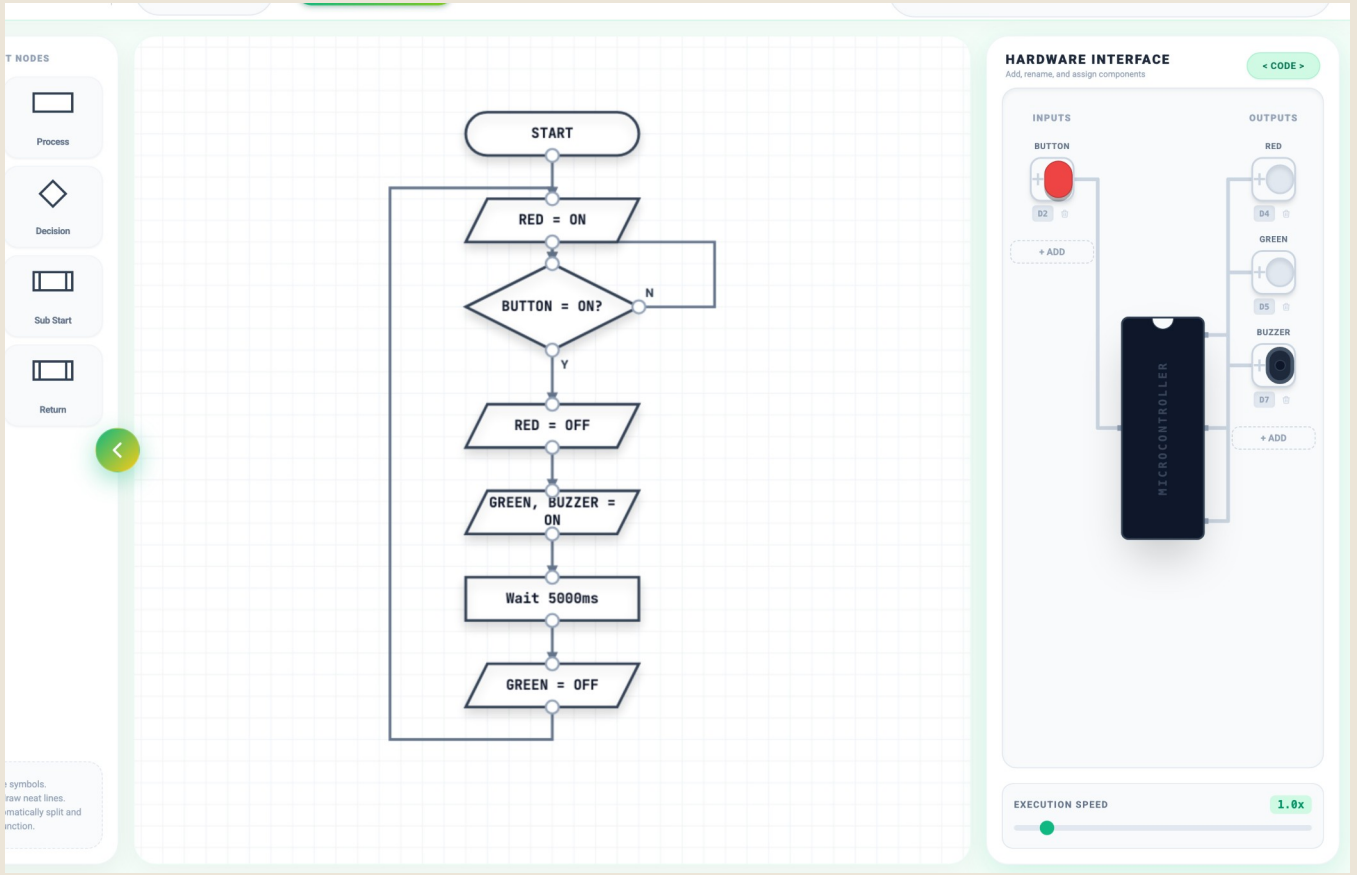
GREEN = OFF

Missing

BUZZER = OFF

So

the beeper never stops



Say What You Saw

THE MISTAKE

- It didn't work.
- Fixed it.
- Works now.

THE CORRECT ANSWER

- Beeper stayed on after 5 s.
- Add pin 7 off after pin 5 off.
- Re-test: beeper stops at 5 s.

Final Version: Save It

After the re-test

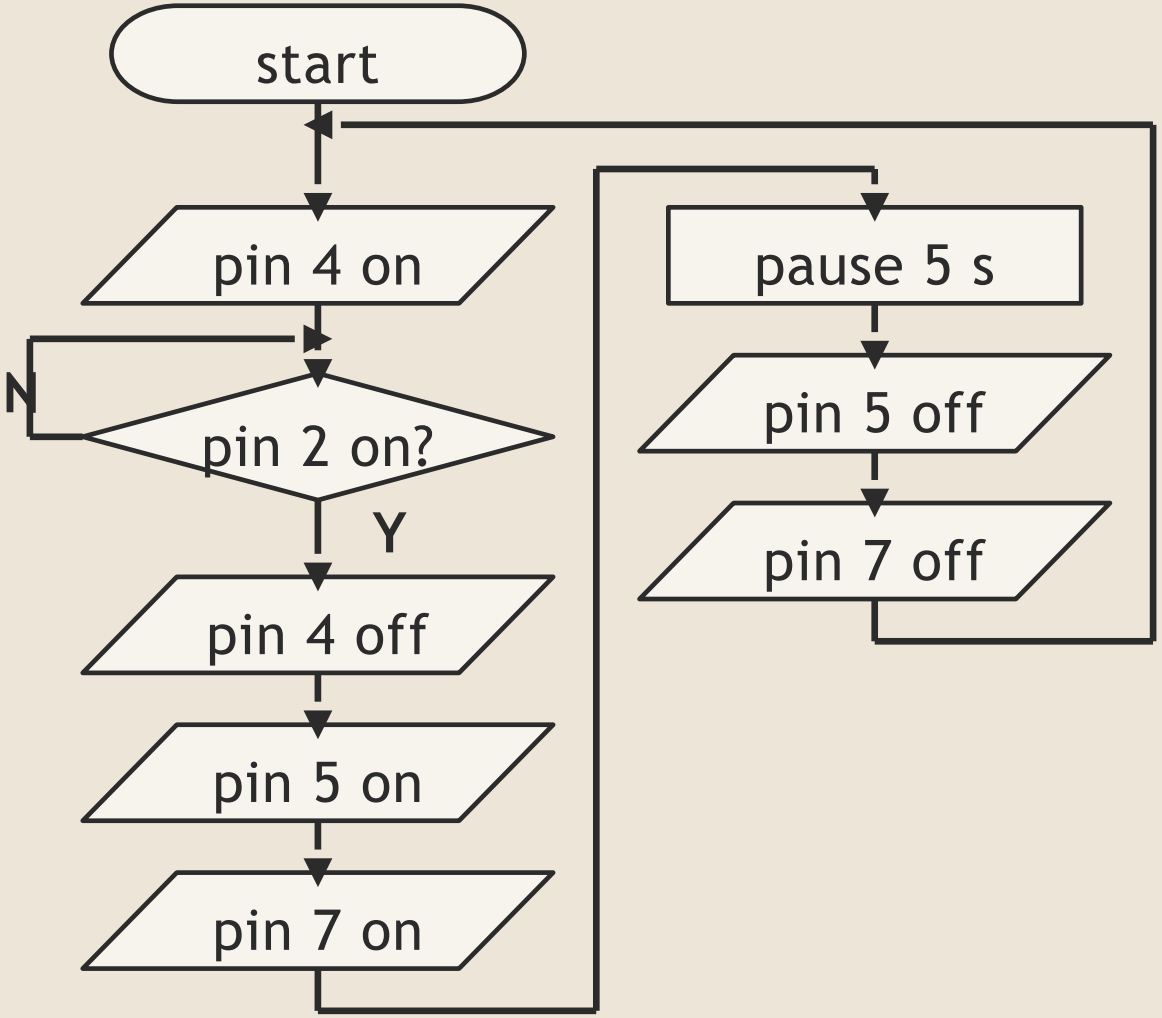
save the final version

Give it

a new name:
crossing_FINAL

It shows

every amendment, all tests passed



Model Answer: Justify an Improvement (2 marks)

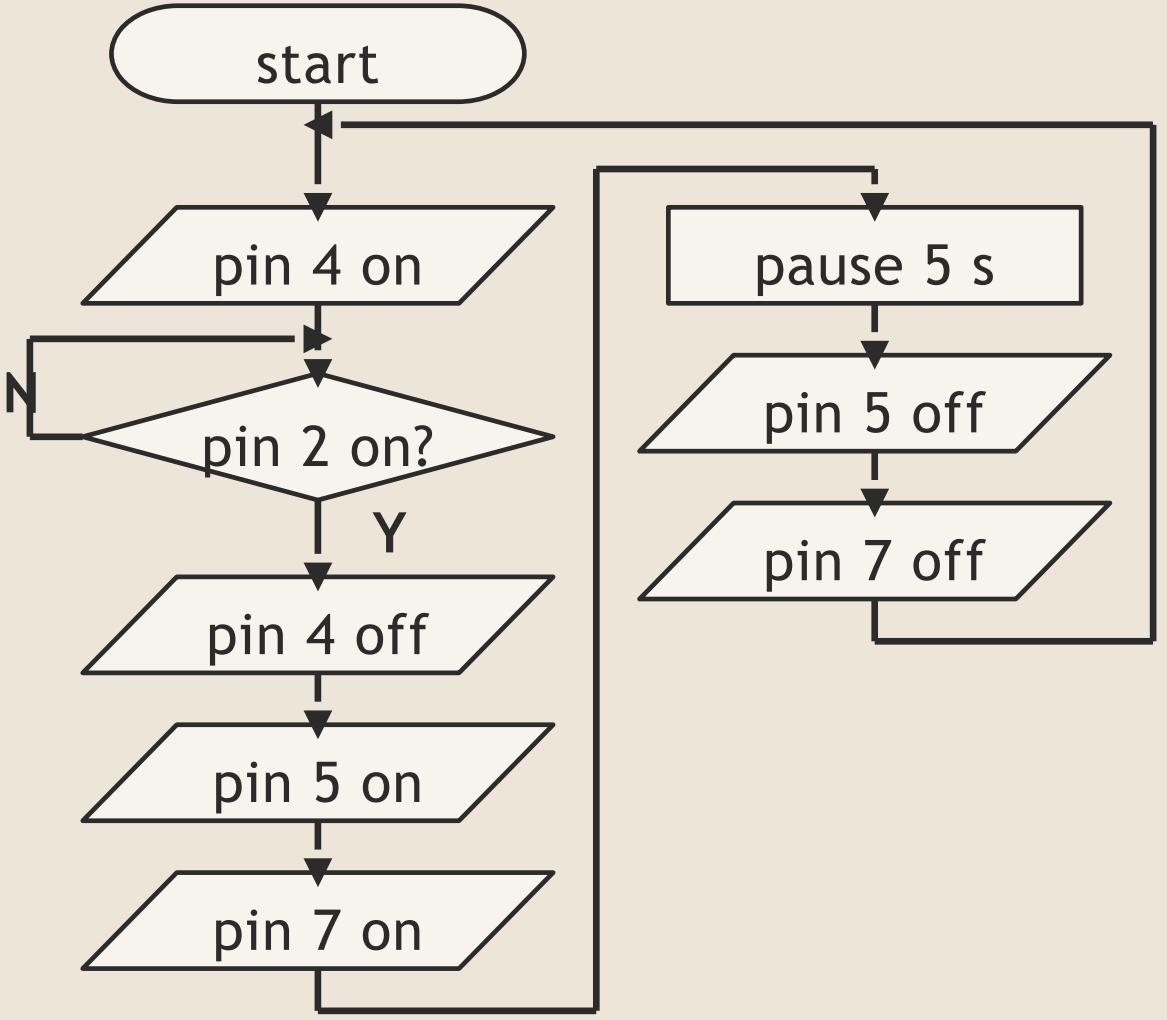
Suggest one improvement to the crossing. Justify it.

Improvement (1 mark)

Make the beeper pulse on and off

Justification (1 mark)

so blind pedestrians know when to cross.



Your Turn: Justify It



1. 'Make it better.' Marks?
2. 'Add a timer.' Marks?
3. Justify a longer green man.

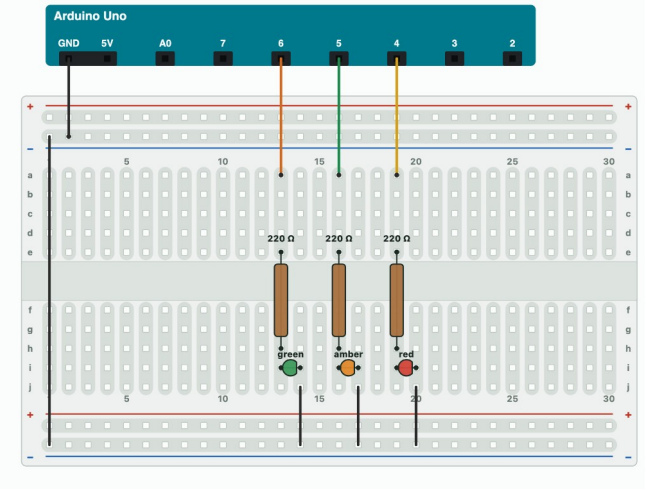
0: no improvement, no reason

1 at most: no justification

So slower walkers cross safely

Test It: Your Traffic Lights

Practical sheet 3



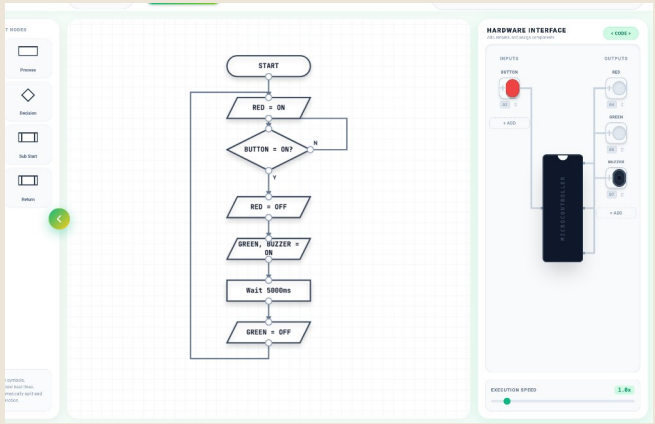
Write 3 tests for your traffic lights.
Run them. Fill in every column.

SAFETY USB power only. Unplug before you move a wire.

12 min

Test It: The Faulty Crossing

Practical sheet 3



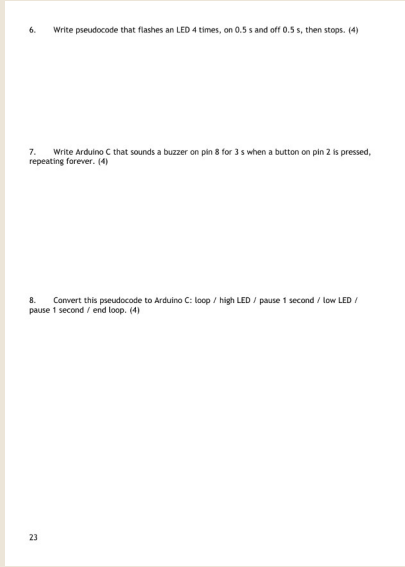
NoSyntaxSim, then Yenka. Test all 3 jobs. Fill the table. Fix it. Re-test.

15 min

PRACTICE: Programming in Real Life

Booklet page 23

Q7 and Q8, on your own.
Then test Q7 on your Uno.



6. Write pseudocode that flashes an LED 4 times, on 0.5 s and off 0.5 s, then stops. (4)

7. Write Arduino C that sounds a buzzer on pin 8 for 3 s when a button on pin 2 is pressed, repeating forever. (4)

8. Convert this pseudocode to Arduino C: loop / high LED / pause 1 second / low LED / pause 1 second / end loop. (4)

23

12 min

Programming in Real Life: Answers



1. Q7 setup()

```
pinMode(2, INPUT);
pinMode(8, OUTPUT);
```

2. Q7 loop()

```
if (digitalRead(2) == HIGH) …
```

3. Q7 inside the if

```
pin 8 HIGH,
delay(3000), LOW
```

Programming in Real Life: Answers



4 Q8 setup()

pinMode for the LED's pin

5 Q8 loop()

HIGH, delay(1000), LOW, delay(1000)

Plenary: Part 3



1	One test for each…	Job the program does
2	'It didn't work' is weak. Why?	It names no part and no result
3	After a change you must…	Re-test
4	Which column says what you saw?	Initial result
5	Change how many things at once?	One

Extension

EXTRA PRACTICE



1. Add amber to the crossing. Where?

   Between green off and red on

2. Make the beeper pulse. How?

   On, pause, off, pause, in a fixed loop

3. The button is never read. Fault?

   Its pin is not pin 2 in the code

Look back — by the end of this part…



1. I can write a test for each job a program does
2. I can say what happened, not just 'it failed'
3. I can plan a change and test it again

LOOKING AHEAD

Booklet 4a: Logic and Digital Control

From programs to logic gates.

Course Specification

Digital electronic control systems

- using suitable commands (high, low, for…next, if…then, pause, end (or their equivalents)) to design programs to solve simple control problems, involving time delays and continuous and fixed loops

Credits (1 of 2)

The big question (An Arduino Uno): Photo: JrawX, CC0, via Wikimedia Commons

From Sim to Board (Uno photo): Photo: JrawX, CC0, via Wikimedia Commons

From Sim to Board (NoSyntaxSim): NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

What to Collect (Uno photo): Photo: JrawX, CC0, via Wikimedia Commons

Breadboard and wiring drawings (all slides): Breadboard and wiring drawings by R Stewart (Stripboard Builder engine).

Watch Out (course specification): © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.9 (question paper).

The Fault, in the Sim: NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

Credits (2 of 2)

Test It: The Faulty Crossing (NoSyntaxSim): NoSyntaxSim V2 screenshot. Simulator by Peter Strain, NoStrainSims (sites.google.com/view/nostrainsim), free and open to all.

PRACTICE: Programming in Real Life (booklet page): Booklet 4b pages by R Stewart.

Course Specification: © Qualifications Scotland (SQA). Course spec v2.0, p.5

R Stewart - October 2026